

**Figure 8.29** (c): When the warm air rises from the surface of the Earth, it adiabatically expands. As a result the water vapor cools and condenses into water droplets forming a cloud.

The equation of state for an adiabatic process is given by

$$PV^{\gamma} = constant \qquad (8.35)$$

Here $\gamma$ is called adiabatic exponent ($\gamma = C_p/C_v$) which depends on the nature of the gas.

The equation (8.35) implies that if the gas goes from an equilibrium state $(P_i, V_i)$ to another equilibrium state $(P_f, V_f)$ adiabatically then it satisfies the relation

$$P_i V_i^{\gamma} = P_f V_f^{\gamma} \qquad (8.36)$$

The PV diagram of an adiabatic expansion and adiabatic compression process are shown in Figure 8.30. The PV diagram for an adiabatic process is also called *adiabat*

Note that the PV diagram for isothermal (Figure 8.25) and adiabatic (Figure 8.30) processes look similar. But actually the adiabatic curve is steeper than isothermal curve.

We can also rewrite the equation (8.35) in terms of T and V. From ideal gas equation, the pressure $P = \frac{\mu RT}{V}$. Substituting this equation in the equation (8.35), we have

$$\frac{\mu RT}{V} V^{\gamma} = \text{constant (or)} \ \frac{T}{V} V^{\gamma} = \frac{constant}{\mu R}$$

Note here that is another constant. So it can be written as

$$TV^{\gamma-1} = constant. \qquad (8.37)$$

The equation (8.37) implies that if the gas goes from an initial equilibrium state $(T_i, V_i)$ to final equilibrium state $(T_f, V_f)$ adiabatically then it satisfies the relation

$$T_i V_i^{\gamma-1} = T_f V_f^{\gamma-1} \qquad (8.38)$$

The equation of state for adiabatic process can also be written in terms of T and P as

$$T^{\gamma} P^{1-\gamma} = constant. \qquad (8.39)$$

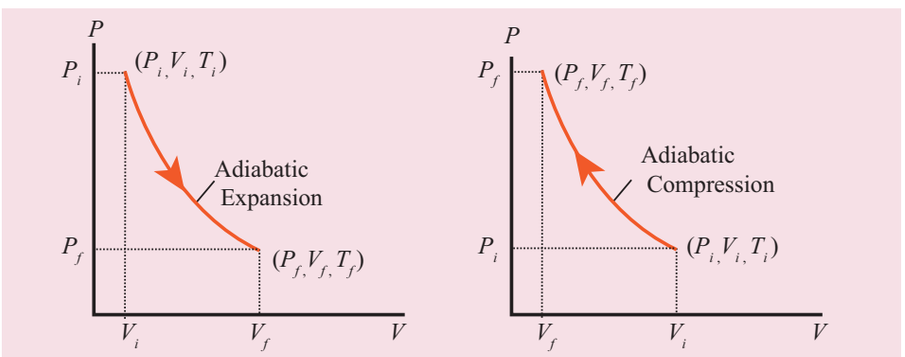



**Figure 8.30** PV diagram for adiabatic expansion and adiabatic compression



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